Assignment - 3.

① (a) 2nd order ODE for classical Harmonic oscillator.

$$\frac{d^2x}{dt^2} + \omega^2 x = 0.$$

General sol^n:

$$x(t) = A\cos\omega t + B\sin\omega t \quad \text{—— (i)}$$

$$\Rightarrow \frac{dx}{dt} = -A\omega\sin\omega t + B\omega\cos\omega t.$$

$$\Rightarrow p(t) = -Am\omega\sin\omega t + Bm\omega\cos\omega t. \quad \text{—— (ii)}$$

$t=0$ in (i):

$$\boxed{x_0 = A}$$

$t=0$ in (ii):

$$\boxed{\frac{p_0}{m\omega} = B}$$

$$\boxed{x(t) = x_0\cos\omega t + \frac{p_0}{m\omega}\sin\omega t} \quad \text{(Ans)}$$

(b) $$|\alpha\rangle = \sum_n c_n|n\rangle$$

$$\Rightarrow \hat{a}|\alpha\rangle = \sum_n c_n\hat{a}|n\rangle.$$

$$\Rightarrow \alpha|\alpha\rangle = \sum_n c_n\sqrt{n}\,|n-1\rangle. \quad [a|n\rangle = \sqrt{n}\,|n-1\rangle]$$

$$\Rightarrow \alpha\sum_n c_n|n\rangle = \sum_n c_n\sqrt{n}\,|n-1\rangle$$

$$\Rightarrow \alpha\sum_n c_{n-1}|n-1\rangle = \sum_n c_n\sqrt{n}\,|n-1\rangle.$$

Comparing both sides:

$$\alpha c_{n-1} = c_n\sqrt{n} \Rightarrow \boxed{c_n = \frac{\alpha\cdot c_{n-1}}{\sqrt{n}}} \quad \text{—— (iii)}$$

$n=1$: $c_1 = \frac{\alpha\cdot c_0}{\sqrt{1}}$

$n=2$: $c_2 = \frac{\alpha\cdot c_1}{\sqrt{2}}$

$n=3$: $c_3 = \frac{\alpha\cdot c_2}{\sqrt{3}}$

$\vdots$

$n=n$: $c_n = \frac{\alpha\cdot c_{n-1}}{\sqrt{n}}$

(iv) —— $$\boxed{c_n = \frac{\alpha^n\cdot c_0}{\sqrt{n!}}}$$

$$|\alpha\rangle = \sum_n c_n|n\rangle.$$

$$\Rightarrow \langle\alpha| = \sum_n c_n^*\langle n|.$$

$$\therefore \langle\alpha|\alpha\rangle = \left(\sum_m c_m^*\langle m|\right)\left(\sum_n c_n^*|n\rangle\right)$$

$$= \sum_n |c_n|^2\langle n|n\rangle = 1.$$

$$\Rightarrow \sum_n |c_n|^2 = 1. \quad [\langle n|n\rangle = \langle\alpha|\alpha\rangle = 1]$$

$$\Rightarrow c_0^2 + c_1^2 + c_2^2 + \cdots\cdots c_n^2 = 1.$$

$$\Rightarrow c_0^2 + \frac{|\alpha|^2 c_0^2}{1} + \frac{|\alpha|^4 c_0^2}{2!} + \frac{|\alpha|^6 c_0^2}{3!} + \cdots = 1.$$

[from (iv)].

$$\Rightarrow c_0^2 \left[1 + |\alpha|^2 + \frac{|\alpha|^4}{2!} + \cdots\right] = 1.$$

$$\Rightarrow c_0^2 \, e^{+|\alpha|^2} = 1 \quad \Rightarrow \boxed{c_0 = e^{-|\alpha|^2/2}} \quad \text{— (v)}$$

$$\boxed{c_n = \frac{\alpha^n}{\sqrt{n!}} e^{-|\alpha|^2/2}} \quad \text{(Ans)} \quad \text{— (vi)}$$

(c) $$|\alpha\rangle = \sum_n c_n |n\rangle$$

$$= \sum_n \frac{\alpha^n}{\sqrt{n!}} e^{-|\alpha|^2/2} \cdot \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle. \quad \text{[from (vi)]}$$

$$= e^{-|\alpha|^2/2} \left(\sum_n \frac{(\alpha a^\dagger)^n}{n!}\right) |0\rangle.$$

$$\boxed{|\alpha\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger} |0\rangle} \quad \text{(Ans)} \quad \text{— (vii)}$$

(d) $$X = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} ; \quad X^\dagger = e^{-|\alpha|^2/2} e^{\alpha^* a}$$

$$X^\dagger X = e^{-|\alpha|^2} \left[e^{\alpha^* a} \cdot e^{\alpha a^\dagger}\right] \quad \text{— (a)}$$

Let $A = \alpha^* a$; $B = \alpha a^\dagger$

$$[A, B] = |\alpha|^2 [a, a^\dagger] = |\alpha|^2 \quad \text{— (viii)}$$

$$\therefore \text{(a)} = e^{\alpha^* a + \alpha a^\dagger + \frac{|\alpha|^2}{2}}$$

$$\therefore X^\dagger X = e^{-\frac{|\alpha|^2}{2}} e^{\alpha^* a + \alpha a^\dagger} \neq I \quad \text{(Proved)}$$

$$\boxed{\begin{aligned} D &= e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{\beta a} \\ D^\dagger &= e^{-|\alpha|^2/2} e^{\beta^* a^\dagger} e^{\alpha^* a} \end{aligned}} \quad \text{— (ix)}$$

$$\therefore D^\dagger D = e^{-|\alpha|^2} e^{\beta^* a^\dagger} \left[e^{\alpha^* a} e^{\alpha a^\dagger}\right] e^{\beta a} \quad \text{— (a)}$$

~~Let~~ ~~$A = \alpha^* a$~~

$$= e^{-\frac{|\alpha|^2}{2}} \left[e^{\beta^* a^\dagger} e^{\alpha^* a + \alpha a^\dagger}\right] \cdot e^{\beta a} \quad \text{— (b)}$$

Let $A = \beta^* a^\dagger$; $B = \alpha^* a + \alpha a^\dagger$

$$[A, B] = [\beta^* a^\dagger, \alpha^* a] + [\beta^* a^\dagger, \alpha a^\dagger].$$

$$= -\beta^* \alpha^* +$$

~~(b) = e^{β^*}~~

(b) $= e^{\beta^* a^\dagger + \alpha^* a + \alpha a^\dagger - \beta^* \alpha^*}$

$$\therefore D^\dagger D = e^{-\frac{|\alpha|^2}{2} - \frac{\beta^*\alpha^*}{2}} \left[e^{\beta^* a^\dagger + \alpha^* a + \alpha a^\dagger} \cdot e^{\beta a} \right] \quad \text{— (c)}$$

Let $A = \beta^* a^\dagger + \alpha^* a + \alpha a^\dagger$; $B = \beta a$

$$[A, B] = |\beta|^2 [a^\dagger, a] + \alpha^* \beta [a, a] + \alpha\beta [a^\dagger, a]$$

$$= -|\beta|^2 - \alpha\beta$$

$$\therefore D^\dagger D = \underbrace{e^{-\frac{|\alpha|^2}{2} + \frac{\beta^*\alpha^*}{2} - \frac{|\beta|^2}{2} - \frac{\alpha\beta}{2}}}_{\text{(d)}} \cdot \underbrace{\exp[\beta^* a^\dagger + \alpha^* a + \alpha a^\dagger + \beta a]}_{\exp[a^\dagger(\beta^* + \alpha) + a(\beta + \alpha^*)]}$$

For $D^\dagger D = \mathbb{I}$;

$$a^\dagger(\beta^* + \alpha) + a(\beta + \alpha^*) = 0 \Rightarrow \beta^* = -\alpha$$

$$\Rightarrow \boxed{\beta = -\alpha^*} \quad \text{(Proved)}$$

Putting $\beta = -\alpha^*$ in (d) :

$$\exp\left(-\frac{1}{2}\left[|\alpha|^2 - |\alpha|^2 - |\alpha|^2 + |\alpha|^2\right]\right) = 1. \quad (*)$$

(e) $$D^\dagger \cancel{a D} = e^{-\frac{|\alpha|^2}{2}} \cdot \left(e^{-\alpha a^\dagger} \cdot e^{\alpha^* a}\right) \quad \text{(e)}$$ [from (VIII)].

Let $A = -\alpha a^\dagger$; $B = \alpha^* a$.

$$[A, B] = -|\alpha|^2 [a^\dagger, a] = |\alpha|^2$$

$$\therefore \boxed{D^\dagger = e^{\frac{|\alpha|^2}{2}} e^{\alpha^* a - \alpha a^\dagger}} \quad \text{— (IX)}$$

$$D = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a}$$

Let $A = \alpha a^\dagger$; $B = -\alpha^* a$.

$$[A, B] = -|\alpha|^2 [a^\dagger, a] = |\alpha|^2$$

$$\therefore \boxed{D = e^{\alpha a^\dagger - \alpha^* a}} \quad \text{— (X)}$$

$$\therefore D^\dagger a D = e^{\alpha^* a - \alpha a^\dagger} \cdot a \cdot e^{\alpha a^\dagger - \alpha^* a}$$

Let $A = \alpha^* a - \alpha a^\dagger \Rightarrow -A = \alpha a^\dagger - \alpha^* a$; $B = \hat{a}$

$$\therefore [A, B] = \alpha^* [a, a] - \alpha [a^\dagger, a] = \alpha.$$

$$[A, [A, B]] = [\alpha^* a - \alpha a^\dagger, \alpha] = 0.$$

$\therefore$ ~~$D^\dagger a D =$~~ ~~Hadamard Lemma~~.

$$\boxed{D^\dagger a D = a + \alpha} \quad \text{— (XI)}$$

Similarly, taking $B = a^\dagger$ & A remaining same:

$$[A, B] = \alpha^* [a, a^\dagger] - \alpha [a^\dagger, a^\dagger] = \alpha^*$$

$$\therefore \boxed{D^\dagger a^\dagger D = a^\dagger + \alpha^*} \quad \text{— (XII)}$$

$$\therefore D^\dagger x D = \sqrt{\frac{\hbar}{2m\omega}}\, D^\dagger (a^\dagger + a) D.$$

$$= \sqrt{\frac{\hbar}{2m\omega}} [D^\dagger a^\dagger D + D^\dagger a D]$$

$$= \sqrt{\frac{\hbar}{2m\omega}} (a^\dagger + \alpha^* + a + \alpha) \quad [\text{from (XI) \& (XII)}]$$

$$= \sqrt{\frac{\hbar}{2m\omega}} (a^\dagger + a) + \sqrt{\frac{\hbar}{2m\omega}} (\alpha^* + \alpha)$$

$$\boxed{D^\dagger x D = \hat{x} + x_0} \quad \text{(Proved)} \quad \text{— (XIII)}$$

$$\therefore D^\dagger p D = i\sqrt{\frac{\hbar m\omega}{2}}\, D^\dagger (a^\dagger - a) D.$$

$$= i\sqrt{\frac{\hbar m\omega}{2}} (a^\dagger + \alpha^* - a - \alpha) \quad [\text{from (XI) \& (XII)}].$$

$$= i\sqrt{\frac{\hbar m\omega}{2}} [(a^\dagger - a) + (\alpha^* - \alpha)].$$

$$\boxed{D^\dagger p D = \hat{p} + p_0} \quad \text{(Proved)} \quad \text{— (XIV)}$$

$$D = e^{\alpha a^\dagger - \alpha^* a} \quad [\text{from (X)}].$$

$$a^\dagger = \frac{1}{\sqrt{2m\hbar\omega}} (-ip + m\omega x)$$

$$a = \frac{1}{\sqrt{2m\hbar\omega}} (ip + m\omega x)$$

$$\therefore \alpha \frac{1}{\sqrt{2m\hbar\omega}} (-\alpha ip + \alpha m\omega x - \alpha^* ip - \alpha^* m\omega x).$$

$$= \frac{1}{\sqrt{2m\hbar\omega}} [m\omega x (\alpha - \alpha^*) - ip (\alpha + \alpha^*)].$$

$$= \frac{1}{\sqrt{2m\hbar\omega}} \left[m\omega x \cdot \sqrt{\frac{2}{\hbar m\omega}}\, i p_0 - ip \cdot \sqrt{\frac{2m\omega}{\hbar}}\, x_0\right] \quad [\text{from (8)}].$$

$$= \frac{i}{\hbar} (x p_0 - p x_0).$$

$$\therefore \boxed{D = \exp\left[\frac{i}{\hbar} (\hat{x} p_0 - \hat{p} x_0)\right]} \quad \text{(Proved)} \quad \text{— (XV)}$$

(8) $$e^{\beta a} |0\rangle = \left(1 + \frac{\beta a}{1!} + \frac{(\beta a)^2}{2!} + \dots\right) |0\rangle = |0\rangle \quad [a|0\rangle = 0].$$

(f) $|\alpha\rangle = D|0\rangle, \Rightarrow \langle\alpha| = \langle 0|D^\dagger$.

$\therefore \langle\alpha|x|\alpha\rangle$.

$= \langle 0|D^\dagger x D|0\rangle = \langle 0|x + x_0|0\rangle$. [from (xiii)]

$= \langle 0|x|0\rangle + \langle 0|x_0|0\rangle$.

$\boxed{\langle\alpha|x|\alpha\rangle = x_0}$ $\quad [\langle 0|x|0\rangle = 0]$.

(Proved)

$\langle\alpha|p|\alpha\rangle = \langle 0|D^\dagger p D|0\rangle$.

$= \langle 0|p + p_0|0\rangle$ [from (xiv)]

$= \langle 0|p|0\rangle + p_0$

$\Rightarrow \boxed{\langle\alpha|p|\alpha\rangle = p_0}$ (Proved)

$|\alpha\rangle = D|0\rangle$.

$\Rightarrow \psi_\alpha(x,0) = \langle x|e^{\frac{i}{\hbar}(xp_0 - px_0)}|0\rangle$ [from (xv)].

$[\langle x|\alpha\rangle = \psi_\alpha(x,0)]$

$= e^{\frac{i}{\hbar}xp_0}\langle x|[e^{-\frac{ipx_0}{\hbar}}]|0\rangle$.

$e^{\frac{i}{\hbar}px_0}|x\rangle = |x+x_0\rangle \Rightarrow \langle x|e^{-\frac{i}{\hbar}px_0} = \langle x-x_0|$.

$\therefore \boxed{\psi_\alpha(x,0) = e^{\frac{ixp_0}{\hbar}}\langle x-x_0|0\rangle = e^{\frac{ixp_0}{\hbar}}\psi_0(x-x_0)}$ (Proved).

(g) $|\alpha(t)\rangle = e^{-iHt/\hbar}|\alpha\rangle$.

$= e^{-iHt/\hbar}\, e^{-|\alpha|^2/2}\, e^{\alpha a^\dagger}|0\rangle = \sum \frac{\alpha^n}{\sqrt{n!}} e^{-|\alpha|^2/2}|n\rangle$.

$= e^{-|\alpha|^2/2}\sum \frac{\alpha^n}{\sqrt{n!}} e^{-iHt/\hbar}|n\rangle$.

$= e^{-|\alpha|^2/2}\sum \frac{\alpha^n}{\sqrt{n!}} e^{-iE_nt/\hbar}|n\rangle$.

$= (\quad)\sum \frac{\alpha^n}{\sqrt{n!}} e^{-i(n+\frac{1}{2})\omega t}|n\rangle$.

$= [(e^{-|\alpha|^2/2})\cdot \sum_n \frac{(\alpha e^{-i\omega t})^n}{\sqrt{n!}}|n\rangle]\cdot e^{-\frac{i\omega t}{2}}$

$\boxed{|\alpha(t)\rangle = e^{-i\omega t/2}\cdot|\alpha e^{-i\omega t}\rangle}$ (Proved)

From (xiii): $D^\dagger x D = \hat{x} + x_0$; $x_0 = \sqrt{\frac{\hbar}{2m\omega}}(\alpha + \alpha^*)$.

Replace $\alpha \longrightarrow \alpha e^{-i\omega t}$:

$$x_0(t) = \sqrt{\frac{\hbar}{2m\omega}}\left(\alpha e^{-i\omega t} + \alpha^* e^{i\omega t}\right).$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left(\alpha \cos\omega t - i\alpha \sin\omega t + \alpha^* \cos\omega t + i\alpha^* \sin\omega t\right),$$

$$x_0(t) = \sqrt{\frac{\hbar}{2m\omega}}\left[\cos\omega t\,(\alpha+\alpha^*) + i\sin\omega t\,(\alpha^* - \alpha)\right].$$

$$\boxed{x_0(t)\ \cancel{x_0(t)} = x_0 \cos\omega t \cancel{-y} + \frac{p_0}{m\omega}\sin\omega t} \quad \text{— (xvi)} \quad \text{(Proved)}$$

~~From (xiv) : $D^\dagger p D = p + p_0$; $p_0 = i\sqrt{\frac{\hbar m\omega}{2}}(\alpha^* - \alpha)$.~~

~~$p_0 \rightarrow$ Replace $\alpha \rightarrow \alpha e^{-i\omega t}$;~~

$$\boxed{p_0(t) = -m\omega x_0 \sin\omega t + p_0 \cos\omega t}, \quad \left[p_0(t) = m\frac{dx_0(t)}{dt}\right].$$

(Proved)

(h) $|\alpha\rangle = D(\alpha)|0\rangle$

$$\Rightarrow |\alpha(t)\rangle = D(\alpha e^{-i\omega t})|0\rangle.$$

$$= e^{-|\alpha|^2/2} e^{\alpha a e^{-i\omega t}} \text{ at } \cancel{e^{-x^*}} |0\rangle.$$

$$= e^{-|\alpha|^2/2} e^{(\alpha e^{-i\omega t})a^\dagger}|0\rangle.$$

$$= |\alpha e^{-i\omega t}\rangle.$$

$$\therefore \psi_\alpha(x,t) = \langle x|\alpha(t)\rangle = \langle x|D(\alpha e^{-i\omega t})|0\rangle.$$

$$= \langle x|\exp\left[\frac{i}{\hbar}(x p_0(t) - p_0 x_0(t))\right]|0\rangle$$

$$= \exp\left(\frac{i}{\hbar}x p_0(t)\right)\langle x|e^{-\frac{i p x_0(t)}{\hbar}}|0\rangle.$$

$$\boxed{\psi_\alpha(x,t) = e^{\frac{i x p_0(t)}{\hbar}}\psi_0(x - x_0(t))}. \quad \text{(Proved)}$$

(3) $\hat{H} = -\omega \hat{S}_z$; $\omega = \mu B$.

(a) $\cancel{[A, H]} = \frac{i}{\hbar}$ $\boxed{\frac{dA}{dt} = \frac{1}{i\hbar}[A, H] + \frac{\partial A}{\partial t}}$ — Heisenberg's EOM.

Taking $A = S_x$:

$$\frac{dS_x(t)}{dt} = \frac{-\omega}{i\hbar}[S_x, S_z] = \omega S_y(t) = \mu B S_y(t). \quad \text{— (i)}$$

Taking $A = S_y$:

$$\frac{dS_y(t)}{dt} = \frac{-\omega}{i\hbar}[S_y, S_z] = -\omega S_x(t) = -\mu B S_x(t). \quad \text{— (ii)}$$

Taking $A = S_z$: $\boxed{\frac{dS_z(t)}{dt}} = 0$ — (iii) (Proved)

(b) Differentiating (iii) : $\frac{d^2 S_z}{dt^2} = 0$.

$$\Rightarrow \boxed{S_z(t) = S_z(0)} \quad \text{— (iv)}$$

(Proved)

Differentiating (ii) :

$$\frac{d^2 S_y}{dt^2} = -\omega \frac{dS_x}{dt} = -\omega^2 S_y$$

$$\Rightarrow \frac{d^2 S_y}{dt^2} + \omega^2 S_y = 0.$$

General Sol^n : $S_y(t) = A \cos \omega t + B \sin \omega t$ — (v)

$$\Rightarrow \frac{dS_y(t)}{dt} = -A\omega \sin \omega t + B\omega \cos \omega t$$

[from (ii)] $\Rightarrow$ $S_x(t) = A \sin \omega t - B \cos \omega t$ — (vi)

@ $t=0$: $S_y(t) = A = S_y(0)$

$S_x(t) = -B = S_x(0)$.

$$\therefore \left.\begin{aligned} S_y(t) &= S_y(0) \cos \omega t - S_x(0) \sin \omega t \\ S_x(t) &= S_y(0) \sin \omega t + S_x(0) \cos \omega t \end{aligned}\right] \text{(Proved)}$$

(5) $H = \omega \hat{S}_y$

Let $|+y\rangle$ & $|-y\rangle$ be the eigenstates of $\hat{S}_y$

$\therefore S_y |+y\rangle = \frac{\hbar}{2} \sigma_y |+y\rangle = \frac{\hbar}{2} |+y\rangle$.

$$H|+y\rangle = \omega S_y |+y\rangle = \frac{\hbar\omega}{2} |+y\rangle = E_+ |+y\rangle \quad \text{— (a)}$$

$$H|-y\rangle = -\frac{\hbar\omega}{2} |-y\rangle = E_- |-y\rangle. \quad \text{— (b)}$$

$$|\psi(0)\rangle = |\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \quad \text{— (i)}$$

$$|+y\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix} \quad \text{— (ii)}$$

$$|-y\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix} \quad \text{— (iii)}$$

S_z basis

$\therefore$ Let $|\uparrow_x\rangle = c_+ |+y\rangle + c_- |-y\rangle$.

$$\Rightarrow c_+ = \langle +y|\uparrow_x\rangle = \frac{1}{2}\begin{pmatrix}1 & -i\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} \quad \text{[from (i) \& (ii)]}$$

$$\boxed{c_+ = \frac{1-i}{2}}$$

$$\Rightarrow c_- = \langle -y|\uparrow_x\rangle = \frac{1}{2}\begin{pmatrix}1 & +i\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} \quad \text{[from (i) \& (iii)]}$$

$$\boxed{c_- = \frac{1+i}{2}}$$

$\therefore |\uparrow_x\rangle = \left(\frac{1+i}{2}\right)|+y\rangle + \left(\frac{1+i}{2}\right)|-y\rangle = |\psi(0)\rangle$

$|+y,t\rangle = e^{-iE_+t/\hbar}|+y(0)\rangle = e^{-i\omega t/2}|+y\rangle$ [from (a) & (b)]

$|-y,t\rangle = e^{-iE_-t/\hbar}|-y(0)\rangle = e^{+i\omega t/2}|-y\rangle$

(Time evolution of stationary states). $|+y\rangle$ & $|-y\rangle$ $|+y\rangle$ & $|-y\rangle$ are stationary " as they are eigenstates of the Hamiltonian).

$\therefore |\psi(t)\rangle = \left(\frac{1+i}{2}\right)e^{-i\omega t/2}|+y\rangle + \left(\frac{1+i}{2}\right)e^{i\omega t/2}|-y\rangle$

$P(t) = |\langle\uparrow|\psi(t)\rangle|^2$

$= \left|\left(\frac{1+i}{2}\right)e^{-i\omega t/2}\cdot\frac{1}{\sqrt{2}} + \left(\frac{1+i}{2}\right)e^{i\omega t/2}\cdot\frac{1}{\sqrt{2}}\right|^2$

$= \frac{1}{2}\left|\frac{(e^{-i\omega t/2}+e^{i\omega t/2})}{2} + i\frac{(e^{i\omega t/2}-e^{-i\omega t/2})}{2}\right|^2$

$P(t) = \frac{1}{2}|\cos\theta + \sin\theta|^2 \quad ; \quad \theta = \frac{\omega t}{2}$

$\boxed{P(t) = \frac{1}{2}(1+\sin 2\theta)}$ (Ans)

for $P(t) = 1 \Rightarrow \sin 2\theta = +1$

$\Rightarrow 2\theta = +\frac{3\pi}{2} + 2\pi n$

$\Rightarrow \frac{\omega t}{2} = +\frac{3\pi}{4} + \pi n \Rightarrow \boxed{t = \frac{3\pi}{2\omega}}$

$\Rightarrow \boxed{t = \frac{1}{\omega}\left(\frac{3\pi}{2} + 2\pi n\right)}$ (Ans) $n \in \mathbb{Z}$

(5) $J = J_1 + J_2$

(a) $J = 2$

$|2,2\rangle = |1,1;1,1\rangle = |1,1\rangle \otimes |1,1\rangle$ — (ix)

$\therefore J_-|2,2\rangle = (J_{1-} + J_{2-})|1,1;1,1\rangle$

$\Rightarrow 2\hbar|2,1\rangle = \sqrt{2}\hbar|1,0;1,1\rangle + \sqrt{2}\hbar|1,1;1,0\rangle$ — (i)

$\Rightarrow |2,1\rangle = \frac{1}{\sqrt{2}}(|1,0;1,1\rangle + |1,1;1,0\rangle)$

Acting with J_-:

$J_-|2,1\rangle = \frac{1}{\sqrt{2}}(J_{1-} + J_{2-})(\quad)$

$\Rightarrow \sqrt{6}\hbar|2,0\rangle = \frac{1}{\sqrt{2}}(\sqrt{2}\hbar|1,-1;1,1\rangle + \sqrt{2}\hbar|1,0;1,0\rangle + \sqrt{2}\hbar|1,0;1,0\rangle + \sqrt{2}\hbar|1,1;1,-1\rangle)$

$\Rightarrow |2,0\rangle = \frac{1}{\sqrt{6}}|1,-1;1,1\rangle + \sqrt{\frac{2}{3}}|1,0;1,0\rangle + \frac{1}{\sqrt{6}}|1,1;1,-1\rangle$ — (ii)

~~Acting with~~ J_- Using property of CG coefficients:

~~$\Rightarrow J=|2,0\rangle \sqrt{6}\hbar\, |2,-1\rangle = \frac{1}{\sqrt{6}}$~~ [

$j_1 = 1 \; ; \; j_2 = 1 \; ; \; j = 2$ $\quad ; (-1)^{j_1+j_2-j} = 1$

~~$|j,m\rangle = (-1)^{j_1+j_2-j} (\,|j_1,m_1 ; j_2,m_2$~~

$$|2,-1\rangle = \frac{1}{\sqrt{2}} \left(|1,0 ; 1,-1\rangle + |1,-1 ; 1,0\rangle \right) \quad \text{— (iii)}$$

Again using property of CG coefficients for $j_1 = 1, j_2 = 1, j = 2$

$(-1)^{j_1+j_2-j} = 1.$

$$|2,-2\rangle = |1,-1 ; 1,-1\rangle \quad \text{— (iv)}$$

(ii) $\underline{J=1}$:

$$|1,1\rangle = c_1 |1,0 ; 1,1\rangle + c_2 |1,1 ; 1,0\rangle \; ; \; c_1^2 + c_2^2 = 1. \quad \text{— (a)}$$

$$\therefore \langle 2,1|1,1\rangle = \frac{c_1}{\sqrt{2}} + \frac{c_2}{\sqrt{2}} = 0 \quad [\text{from (i)}]$$

$$\Rightarrow c_1 = -c_2 \quad \text{— (b)}$$

From (a) & (b) : $\boxed{c_1 = \frac{1}{\sqrt{2}} \; ; \; c_2 = -\frac{1}{\sqrt{2}}}$

$$|1,1\rangle = \frac{1}{\sqrt{2}} \left(|1,0 ; 1,1\rangle - |1,1 ; 1,0\rangle \right) \quad \text{— (v)}$$

Acting with J_- :

$$\sqrt{2}\hbar\, |1,0\rangle = \frac{1}{\sqrt{2}} \left(\sqrt{2}\hbar\, |1,-1 ; 1,1\rangle - \sqrt{2}\hbar\, |1,1 ; 1,-1\rangle \right)$$

$$\Rightarrow |1,0\rangle = \frac{1}{\sqrt{2}} \left(|1,-1 ; 1,1\rangle - |1,1 ; 1,-1\rangle \right) \quad \text{— (vi)}$$

CG coefficient property : $j_1 = 1 \; ; \; j_2 = 1 \; ; \; j = 1.$

$$|1,-1\rangle = -\frac{1}{\sqrt{2}} \left(|1,0 ; 1,-1\rangle - |1,-1 ; 1,0\rangle \right)$$

$$|1,-1\rangle = \frac{1}{\sqrt{2}} \left(|1,-1 ; 1,0\rangle - |1,0 ; 1,-1\rangle \right) \quad \text{— (vii)}$$

(iii) $\underline{J=0}$:

$$|0,0\rangle = c_1 |1,-1 ; 1,1\rangle + c_2 |1,0 ; 1,0\rangle + c_3 |1,1 ; 1,-1\rangle.$$

$$c_1^2 + c_2^2 + c_3^2 = 1. \quad \text{— (b)}$$

$$\langle 2,0|0,0\rangle = \frac{c_1}{\sqrt{6}} + \sqrt{\frac{2}{3}}\, c_2 + \frac{c_3}{\sqrt{6}} = 0 \quad [\text{from (ii)}]$$

$$\langle 1,0|0,0\rangle = \frac{c_1}{\sqrt{2}} - \frac{c_3}{\sqrt{2}} = 0 \Rightarrow c_1 = c_3 \quad [\text{from (vi)}].$$

$$\therefore c_1 + c_2 = 0 \Rightarrow c_1 = -c_2$$

$$\therefore \boxed{c_1 = -c_2 = c_3 = \frac{1}{\sqrt{3}}}$$

$\therefore |0,0\rangle = \frac{1}{\sqrt{3}}|1,-1;1,0\rangle + -\frac{1}{\sqrt{3}}|1,0;1,0\rangle + \frac{1}{\sqrt{3}}|1,1;1,-1\rangle$ — (viii)

① — (viii) (ix) are the coefficients for all the states for all the sectors.

② (a) For $x \neq \frac{a}{2}$.

$$-\frac{\hbar^2}{2m}\psi'' = E\psi. \Rightarrow \psi'' + \frac{2mE}{\hbar^2}\psi = 0.$$

$$\Rightarrow \boxed{\psi'' + k^2\psi = 0} \; ; \; k^2 = \frac{2mE}{\hbar^2}$$

: General solution:

$$\psi(x) = \begin{cases} A_1 \sin kx + B_1 \cos kx & ; \; 0 < x < \frac{a}{2} \\ A_2 \sin kx + B_2 \cos kx & ; \; \frac{a}{2} < x < a \\ 0 & ; \; \text{otherwise} \end{cases}$$

(i) Continuity @ $x = \frac{a}{2}$:

$$A_1 \sin\frac{ka}{2} + B_1 \cos\frac{ka}{2} = A_2 \sin\frac{ka}{2} + B_2 \cos\frac{ka}{2}$$

$$\Rightarrow \sin\left(\frac{ka}{2}\right)(A_1 - A_2) = \cos\left(\frac{ka}{2}\right)(B_2 - B_1) \quad \text{— (i)}$$

(ii) Derivative jump @ $x = \frac{a}{2}$:

$$\int_{\frac{a}{2}-\epsilon}^{\frac{a}{2}+\epsilon} -\frac{\hbar^2}{2m}\psi'' + \int V_0 \delta\left(x - \frac{a}{2}\right)\psi = \int E\psi.$$

$$\Rightarrow -\frac{\hbar^2}{2m}\psi'\Big|_{\frac{a}{2}-\epsilon}^{\frac{a}{2}+\epsilon} + V_0\psi\left(\frac{a}{2}\right) = 0 \qquad [\epsilon \to 0]$$

$$\Rightarrow \boxed{\psi'\left(\frac{a}{2}^+\right) - \psi'\left(\frac{a}{2}^-\right) = \frac{2mV_0}{\hbar^2}\psi\left(\frac{a}{2}\right)} \quad \text{— (ii)}$$

$$\therefore \psi'\left(\frac{a}{2}^+\right) = kA_2 \cos kx - kB_2 \sin kx \Big|_{\frac{a}{2}}$$

$$= k\left[A_2 \cos\frac{ka}{2} - B_2 \sin\frac{ka}{2}\right]$$

$$\psi'\left(\frac{a}{2}^-\right) = k\left[A_1 \cos\frac{ka}{2} - B_1 \sin\frac{ka}{2}\right]$$

$$\therefore \frac{2mE}{\hbar^2}\left[\cos\frac{ka}{2}(A_2 - A_1) - \sin\left(\frac{ka}{2}\right)(B_2 - B_1)\right] = \frac{2mV_0}{\hbar^2}\psi\left(\frac{a}{2}\right)$$

$$\Rightarrow E\left[-\frac{(A_2 - A_1)^2}{(B_2 - B_1)} - (B_2 - B_1)^2\right]\sin\frac{ka}{2} = V_0$$

(iii) Boundary conditions :

(i) $\psi(0) = 0 \Rightarrow \boxed{B_1 = 0}$ — (iii)

$\therefore \psi(a) = 0 \Rightarrow B_2 \cos ka + A_2 \sin ka = 0$ (1)

$\Rightarrow \boxed{B_2 = -\frac{A_2 \sin ka}{\cos ka}}$ — (iv)

Substituting (iv) in (1) :

$A_1 \sin ka (A_1 - A_2) = B_2 \cos\left(\frac{ka}{2}\right)$

$\Rightarrow$

$$\psi(x) = \begin{cases} A_1 \sin kx & ; x < \frac{a}{2} \\ A_2 \sin kx - A_2 \cancel{\sin ka} \tan ka \cos kx & ; x > \frac{a}{2} \end{cases}$$

$$\psi'\left(\frac{a}{2}^+\right) = kA_2 \cos\left(\frac{ka}{2}\right) + kA_2 \tan(ka) \sin\left(\frac{ka}{2}\right)$$

$$\psi'\left(\frac{a}{2}^+\right) = kA_2 \left[\cos\left(\frac{ka}{2}\right) + \tan(ka) \sin\left(\frac{ka}{2}\right)\right].$$

$$\therefore \psi'\left(\frac{a}{2}^-\right) = kA_1 \cos\left(\frac{ka}{2}\right)$$

From (ii): $kA_2 \left[\cos\left(\frac{ka}{2}\right) + \tan(ka) \sin\left(\frac{ka}{2}\right)\right] - kA_1 \cos\left(\frac{ka}{2}\right)$

$$= \frac{2m V_0}{\hbar^2} A_1 \sin\left(\frac{ka}{2}\right) \quad \text{— (v)}$$

ii) Continuity @ $x = \frac{a}{2}$:

$$A_1 \sin\left(\frac{ka}{2}\right) = A_2 \sin\left(\frac{ka}{2}\right) - A_2 \tan ka \cos\left(\frac{ka}{2}\right)$$

$$\Rightarrow A_1 = A_2 \left(1 - \frac{\tan ka \cos\left(\frac{ka}{2}\right)}{\sin\left(\frac{ka}{2}\right)}\right).$$

$$= A_2 \left[1 - \frac{2 \sin\left(\frac{ka}{2}\right) \cos^2\left(\frac{ka}{2}\right)}{\sin\left(\frac{ka}{2}\right) \cos ka}\right].$$

$$= \frac{A_2}{\cos ka}\left[\cos ka - 2\cos^2\left(\frac{ka}{2}\right)\right] = -\frac{A_2 \cancel{\cos ka}}{\cos ka}$$

$$\Rightarrow \boxed{-A_1 \cos ka = A_2} \quad \text{— (vi)}$$

$$\therefore \psi(x) = \begin{cases} A_1 \sin kx & ; x < \frac{a}{2} \\ -A_1 \cos ka \sin kx + A_1 \sin ka \cos kx & ; x > \frac{a}{2} \end{cases}$$

$$\psi(x) = \begin{cases} A_1 \sin kx & ; x < \frac{a}{2} \\ -A_1 \sin[k(x-a)] & ; x > \frac{a}{2} \end{cases} \quad \text{[Ans]}$$

$N_1 = A_1$; $N_2 = -A_1$.

(b) From (v) & (vi):

$$kA_2 \left[-\cancel{A_1}\left[\cos ka \cos\left(\frac{ka}{2}\right) + \sin ka \sin\left(\frac{ka}{2}\right)\right]\right] - k\cancel{A_1} \cos\left(\frac{ka}{2}\right) = \frac{2m V_0 \cancel{A_1}}{\hbar^2} \sin\left(\frac{ka}{2}\right)$$

$$\Rightarrow -k\left[\cancel{2}\cos\left(\frac{ka}{2}\right)\right] = \frac{\cancel{2}m V_0}{\hbar^2} \sin\left(\frac{ka}{2}\right)$$

$$\Rightarrow \boxed{\tan\left(\frac{ka}{2}\right) = -\frac{\hbar^2 k}{m V_0}} \quad \text{(Proved)}$$